

UEC630: DEEP LEARNING AND APPLICATIONS				
	L	T	P	Cr
	2	0	2	3.0
Course Objective: To introduce students to fundamentals of deep learning and their applications				
Syllabus Introduction: Introduction to Deep Learning, Deep Supervised Learning, Deep Supervised Learning. Deep Learning Fundamentals: Optimization Methods and Regularization: Gradient Descent, Stochastic Gradient Descent (SGD), Regularization; Neural Network Fundamentals, Convolutional Neural Networks: History of Convolutional Networks, Convolutional Networks and Computer Vision, Audio and Other Domains, Structural Prediction and Natural Language Processing Learning with Memory: Recurrent Neural Network Basics, Advanced Recurrent Neural Networks, Sequential Data Modeling, Embedding Methods for NLP: Unsupervised and Supervised Embeddings, Embedding Methods for NLP: Embeddings for Multi-relational Data, Deep Natural Language Processing Applications: LeNet: Recognizing Handwritten Digits, MiniVGGNet: Going Deeper with CNNs.				
Laboratory Work: Introduction to Python, Development of a neural network, Apply gradient descent algorithm, Explore LeNet, Use pertained models for recognition of MINST/CIFAR-10.				
Course Learning Objectives (CLO) The students will be able to: 1. Explain fundamentals of deep learning 2. Apply optimisation techniques on neural networks. 3. Design and implement CNN architectures 4. Implement and evaluate deep learning models for real-world applications.				
Text Books 1. <i>Deep Learning</i> by Aaron Courville, Ian Goodfellow, and YoshuaBengio 2. <i>Deep Learning with Python</i> by François Chollet 3. <i>Deep Learning: A Practitioner's Approach</i> by Adam Gibson and Josh Patterson				